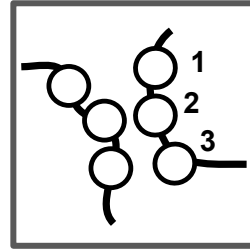


Goal : De Novo Design of Proteins to Neutralize SARS-CoV-2

Amino Acid Type	Amino Acid	1 Letter Code
Hydrophobic: Aliphatic	Glycine	G
	Alanine	A
	Valine	V
	Leucine	L
	Isoleucine	I
	Proline	P
	Methionine	M
Hydrophobic: Aromatic	Phenylalanine	F
	Tyrosine	Y
	Tryptophan	W
Hydrophilic: Polar Uncharged	Serine	S
	Threonine	T
	Cysteine	C
	Asparagine	N
	Glutamine	Q
Hydrophilic: Acidic	Aspartic Acid	D
	Glutamic Acid	E
Hydrophilic: Basic	Arginine	R
	Histidine	H
	Lysine	K

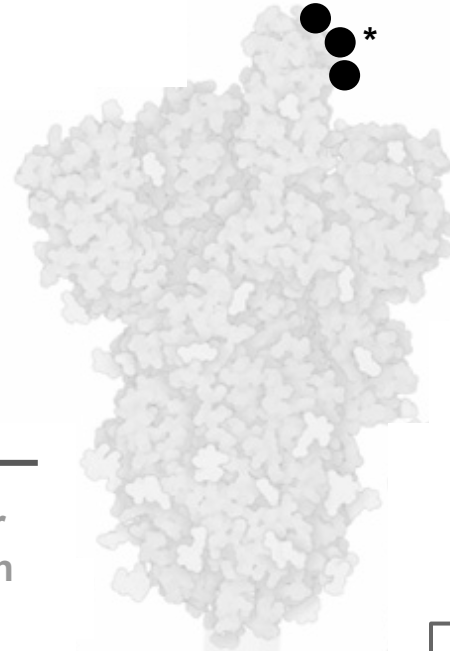


* Binding Interface

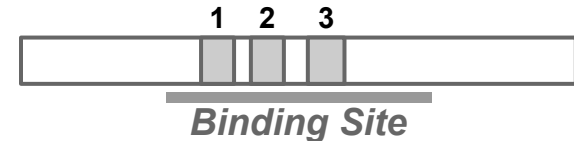
Target

Subcellular Localisation

Scaffold

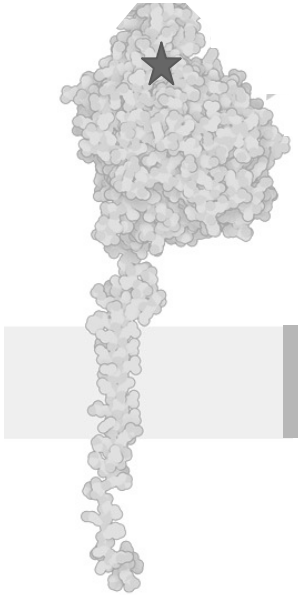


Final Sequence



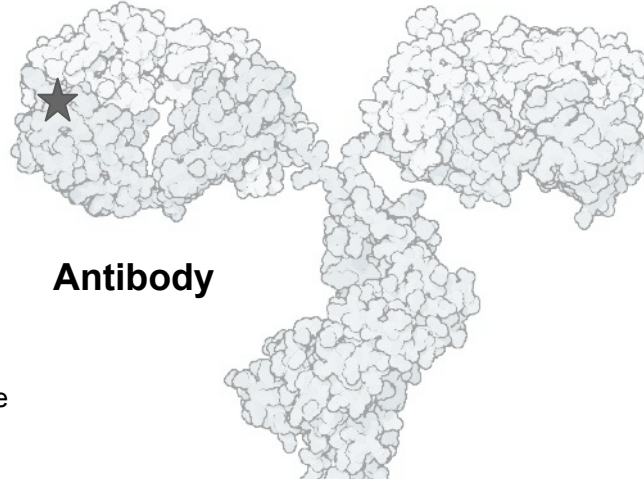
Protein Scaffolds

ACE2 Receptor



Transmembrane
region

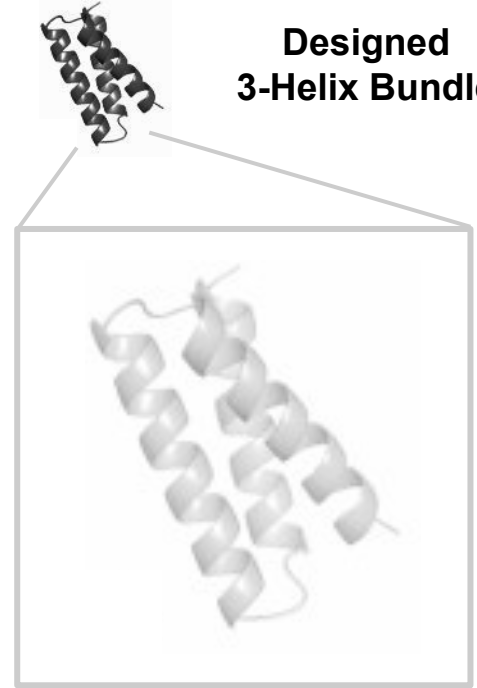
★ Binding domain



Antibody

**If you chose “Generative AI”,
draw your own scaffold!**

**Designed
3-Helix Bundle**



These structures are not to scale or completely accurate, they are only for representation purposes for this activity

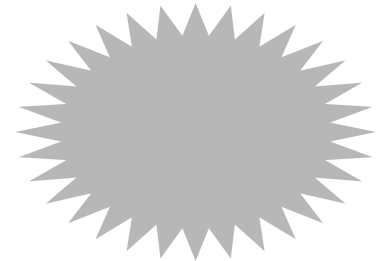
Instructions

- Each team will pull a **"Fate" Card**, which determines the specific starting budget of resources and time units your laboratory must work with.
- The goal of this game is **maximizing both your Novelty and Confidence points** while carefully avoiding bankruptcy by not running out of money and time resources.
- We will handle a **tie on the final leaderboard** by adding any remaining money and time surplus to your final score to determine the absolute winner.
- We will divide each **four-person laboratory** into roles :
 - **Project Leader** - in charge of driving decisions and consulting the team
 - **Lab Manager** - in charge of maintaining the team balance sheet ledger
 - **Experimentalist** - in charge of doing the 'experiment' (filling out worksheet, cutting, and coloring)
 - **Corresponder** - answers any questions we pose during the lecture, defends the team's choices
- I will explain your **available options at the start of each round**, and your team must select at least one option.
- We require your team to actively huddle and try to **estimate how much time and resources** you will lose versus what **point metrics you will gain** before we reveal the true costs of each technique.
- We expect all team members to be actively consulted during every major design decision, but the **Project Leader's choice** will serve as the definitive tie-breaker vote if the group splits.
- We will offer opportunities to score **extra bonus confidence points** at the end of each round by asking 2 to 3 groups to stand up and explain the **pros of their choices**, followed by asking other groups to act as peer reviewers and **critique the cons** of that technique.
- We will award your team **+5 bonus points** if your peer review arguments match the hidden pros and cons on our master slide, and an impressive **+10 points** if you raise a highly valid scientific point that we haven't mentioned
- We will introduce random "Action Cards" in the middle of the game that your laboratory will have to deal with - for good or for worse

Balance Sheet

	Choices	Time (TU)	Resources (RU)	Confidence Points	Novelty Points
Fate Card					
Action Card					
Target Selection					
Scaffold Selection					
Binding Site Optimization					
Binding Site complementarity					
Testing and Validation					
TOTAL :					

Instructor Points



Final Score